

Theorem (Young's inequality). *Let $a, b \geq 0$ and let $p, q > 1$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$. Then*

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q},$$

and equality holds iff either $a = b = 0$ or $a^p = b^q$.

Proof. **Hypotheses and notation.** We are given $a, b \geq 0$ and $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$. Define

$$\lambda := \frac{1}{p}, \quad 1 - \lambda := \frac{1}{q}.$$

From the relation $\frac{1}{p} + \frac{1}{q} = 1$ we obtain $1 - \frac{1}{p} = \frac{1}{q}$.

Reduction to the non-trivial case. If $a = 0$ then the left-hand side of the desired inequality is 0 and the right-hand side is $\frac{0^p}{p} + \frac{b^q}{q} \geq 0$; similarly if $b = 0$. Hence the inequality holds in those cases. Consequently we may assume

$$a > 0, \quad b > 0.$$

Convexity of the exponential function. The function $f(t) = e^t$ is convex on $\mathbb{R}$ because $f''(t) = e^t > 0$ for all t . By Jensen's inequality for a convex function, for any real numbers x, y and any $\lambda \in [0, 1]$,

$$e^{\lambda x + (1-\lambda)y} \leq \lambda e^x + (1-\lambda)e^y. \quad (J)$$

Choice of the arguments. Set

$$x := p \ln a, \quad y := q \ln b.$$

Then, using the definition of λ ,

$$\begin{aligned} \lambda x + (1-\lambda)y &= \frac{1}{p} (p \ln a) + \frac{1}{q} (q \ln b) \\ &= \ln a + \ln b \\ &= \ln(ab). \end{aligned} \quad (1)$$

Apply Jensen's inequality. Substituting x, y and λ into (J) and using (1) we obtain

$$e^{\ln(ab)} \leq \frac{1}{p} e^{p \ln a} + \frac{1}{q} e^{q \ln b}.$$

The left-hand side simplifies to ab , and the right-hand side simplifies using $e^{p \ln a} = a^p$ and $e^{q \ln b} = b^q$:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}. \quad (2)$$

Equality condition. Since e^t is strictly convex, equality in (J) occurs iff the two arguments are equal, i.e. $x = y$, which is equivalent to $p \ln a = q \ln b$.

Exponentiating gives $a^p = b^q$. Together with the trivial case $a = b = 0$ handled above, equality in (2) occurs exactly when $a^p = b^q$ or $a = b = 0$.

Conclusion. For all non-negative real numbers a, b and conjugate exponents $p, q > 1$ satisfying $\frac{1}{p} + \frac{1}{q} = 1$, inequality (2) holds, and equality holds precisely under the condition described above. $\square$